

HOMEWORK 4

NAME:

Due date: Monday

Exercises: 4, 7, (page 34)

Exercises: 9, (page 40)

Exercises: 7, 9, 11, 13 (page 49)

Problem 1. Let F be a field and V be a vector space over F .

- (1) Show that the scalar product of the zero element of F with any vector in V is the zero vector.
- (2) Assume that $\dim_F V < \infty$. Let S be a finite subset which spans V , show that S contains a basis of V , namely, one can obtain a basis of V by deleting several elements from S .
- (3) Assume that $\dim_F V = n$ and $S = \{\alpha_1, \dots, \alpha_n\}$ is a set of n vectors. If S is linearly independent, show that S spans V and thus is a basis of V .
- (4) Assume that $\dim_F V = n$ and $S = \{\alpha_1, \dots, \alpha_n\}$ is a set of n vectors. If S spans V , show that S is linearly independent and thus is a basis of V .

Problem 2. Let F be a field and m, n be positive integers.

- (1) Show that $\dim_F \text{Mat}_{m \times n}(F) = mn$.
- (2) For any $A \in M_{n \times n}(F)$. Show that there is an integer N such that A satisfies a nontrivial polynomial equation

$$A^N + c_{N-1}A^{N-1} + \dots + c_1A + c_0I_n = 0.$$

(Hint: Consider the set $\{I_n, A, A^2, \dots, A^N\} \subset M_{n \times n}(F)$ for a suitable integer N . Then apply Theorem 4 of page 44 and part (1).)

Problem 3. Let F be a field and let n be a positive integer. Let $V = \text{Mat}_{n \times n}(F)$, which is a vector space of dimension n^2 over F by Problem 2. Let $W = \{A \in V \mid \text{tr}(A) = 0\}$.

- (1) Show that W is a subspace of V ;
- (2) Find a subspace U of V such that $U + W = V$.
- (3) Compute $\dim_F(W)$.

Problem 4. In this problem, you are assumed to know polynomials (of one variable/2-variables).

- (1) Let $x(t), y(t)$ be quadratic polynomials with real coefficients (for example, $x(t) = t^2 + t + 1, y(t) = t^2 - 2t - 3$). Show that there is a quadratic polynomial $f(x, y)$ (i.e., $f(x, y)$ is of the form $c_0 + c_1x + c_2y + c_3x^2 + c_4xy + c_5y^2$) such that $f(x(t), y(t))$ is identically zero.
- (2) Let $x(t) = t^2 - 1, y(t) = t^3 - t$. Find a nonzero real polynomial $f(x, y)$ such that $f(x(t), y(t))$ is identically zero.
- (3) Prove that every pair $x(t), y(t)$ of real polynomials satisfies some real polynomial relation $f(x, y) = 0$.

(Hint: Use similar method as Problem 2).

Problem 5. Let α be a real cubic root of 2 (i.e., $\alpha = \sqrt[3]{2}$).

- (1) Show that the set $\{1, \alpha, \alpha^2\}$ are linearly independent over $\mathbb{Q}$, i.e., if $a, b, c \in \mathbb{Q}$ such that $a + b\alpha + c\alpha^2 = 0$, then $a = b = c = 0$.
- (2) Show that the set $F = \{a + b\alpha + c\alpha^2 \mid a, b, c \in \mathbb{Q}\}$ is a field.

(Hint for (1): Proof by contradiction. Assume that $a, b, c \in \mathbb{Z}$ such that $a + b\alpha + c\alpha^2 = 0$ and divide $x^3 - 2$ by $cx^2 + bx + a$.)

Problem 6. Set $\alpha = \sqrt[3]{2}$. Let $F = \{a + b\alpha + c\alpha^2 : a, b, c \in \mathbb{Q}\}$. We know that F is a field and has dimension 3 over $\mathbb{Q}$ from last problem. Thus F^n has dimension $3n$ as a $\mathbb{Q}$ vector space. Choose an ordered basis $\mathcal{B}$ of F^n (as a $\mathbb{Q}$ -vector space). Compute $[v]_{\mathcal{B}}$ for $v = (v_1, \dots, v_n) \in F^n$, where $v_i = a_i + b_i\alpha + c_i\alpha^2$.

You **don't** have to do the next problem.

Problem 7. Let V be a vector space over a field F . A subspace W of V is called proper if $W \neq V$. Show that if F is infinite, V is not the union of finitely many proper subspaces.

(You can find a proof from this [link](#). Please try to fill the details of the proof. You don't have to submit a solution of this problem.)

1. CONSTRUCTION OF NUMBER SYSTEMS

You don't have to do any problems below. But you are advised to, At least, find a book (say, Tao's book) and read the proofs.

Here is a very brief sketch of constructions of number systems. Here is an axiomatic definition of natural number $\mathbb{N}$. The natural numbers are a triple $(\mathbb{N}, 0, S)$, where $\mathbb{N}$ is a set, $0 \in \mathbb{N}$ is an element and $S : \mathbb{N} \rightarrow \mathbb{N}$ is a map such that

- (1) S is injective;
- (2) $0 \notin \text{Im}(S)$;
- (3) If $M \subset \mathbb{N}$ is a subset such that $0 \in M$ and $S(M) \subset M$, then $M = \mathbb{N}$.

The map S is called the successor map (which is just the plus 1 map), 0 is called zero. Part (3) above is the induction principle of $\mathbb{N}$. One then define $1 := S(0), 2 := S(1), \dots$. From S one can define a binary operation (addition) $+: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ inductively in the following way. For a fixed $m \in \mathbb{N}$, we define $m + n$ by induction on n : define $m + 0$ to be m itself. Suppose $m + n$ is defined by some $n \in \mathbb{N}$, then define $m + S(n) := S(m + n)$. One can also define multiplication inductively on $\mathbb{N}$ as follows. For a fixed m . Define $m \cdot 0 = 0$. Let $n \in \mathbb{N}$ suppose that $m \cdot n$ is defined, then define $m \cdot S(n) = m \cdot n + n$. Then one checks the addition and multiplication satisfy all the familiar properties. One can also define an order on $\mathbb{N}$ in the following way. Given $m, n \in \mathbb{N}$, we define $m \geq n$ if there exists an element $k \in \mathbb{N}$ such that $m = n + k$.

The followings are constructions of $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ from $\mathbb{N}$.

Problem 8. Let $X = \mathbb{N} \times \mathbb{N}$ and $R = \{((a, b), (c, d)) \in X \times X \mid a + d = b + c\}$. Check that R is an equivalence relation on X . Let X/R be the set of equivalence classes. On X/R one can define an addition $+: X/R \times X/R \rightarrow X/R$ by

$$\overline{(a, b)} + \overline{(c, d)} = \overline{(a + c, b + d)}.$$

Moreover, one defines the product $\cdot : X/R \times X/R \rightarrow X/R$ by

$$\overline{(a, b)} \cdot \overline{(c, d)} = \overline{(ac + bd, bc + ad)}.$$

Show that the above addition and multiplication are well-defined (namely, independent on the choice of representatives.) Moreover, show that $(X/R, +)$ is an abelian group.

The pair $(X/R, +)$ is just the integers $\mathbb{Z}$. One can define an order on X/R in the following way. For $\overline{(a, b)} \in X/R$, we say $\overline{(a, b)} > 0$ if $a > b$ (note $a, b \in \mathbb{N}$ and the sign $>$ is already defined on $\mathbb{N}$). For $\overline{(a, b)}, \overline{(c, d)} \in X/R$, we say $\overline{(a, b)} > \overline{(c, d)}$ if there exists $\alpha \in X/R, \alpha > 0$ such that $\overline{(a, b)} = \overline{(c, d)} + \alpha$.

Problem 9. Let $Y = \mathbb{Z} \times (\mathbb{Z} - \{0\})$ and $R = \{((a, b), (c, d)) \in Y \times Y \mid ad = bc\}$. Show that R is an equivalence relation on Y . Consider the set of equivalence classes Y/R . This is the set of rational numbers $\mathbb{Q}$.

Define the addition and multiplication on Y/R by yourself and check that it is a field. One also define absolute values on $\mathbb{Q}$ in the usual way. One can easily define an order on $\mathbb{Q}$, which we omit.

The definition of $\mathbb{R}$ is more complicated. A sequence of rational numbers is map $a : \mathbb{N} \rightarrow \mathbb{Q}$. Let a be a sequence of rational numbers. For $n \in \mathbb{N}$, write $a_n := a(n) \in \mathbb{Q}$. Let Z be the set of all sequences of rational numbers, namely, $Z = \{a : \mathbb{N} \rightarrow \mathbb{Q}\}$. Given $a, b \in Z$, we can define $a + b$ and ab

by $(a+b)_n = a_n + b_n$ and $(ab)_n = a_n b_n$. Note that $(Z, +)$ is an abelian group with the zero sequence $\mathbf{0}$ as its additive identity, where $\mathbf{0} : \mathbb{N} \rightarrow \mathbb{Q}$ is the sequence defined by $\mathbf{0}_n = 0$ for any $n \in \mathbb{N}$. But $(Z - \{0\}, \cdot)$ is an abelian group. Actually, the sequence $\mathbf{1} : \mathbb{N} \rightarrow \mathbb{Q}$ defined by $\mathbf{1}_n = 1$ for any $n \in \mathbb{N}$ is a multiplicative identity. But not every nonzero sequence has an additive inverse. For example, consider the sequence $a : \mathbb{N} \rightarrow \mathbb{Q}$ defined by $a_0 = 0$ and $a_n = 1$ for all $n > 0$. Then a is not a zero sequence but a is not invertible. To remedy this, we define Cauchy sequence. A sequence $a : \mathbb{N} \rightarrow \mathbb{Q}$ is called Cauchy if for any $\epsilon > 0, \epsilon \in \mathbb{Q}$, there exists an integer N such that $|a_n - a_m| < \epsilon$ for all $m, n > N$. Let $\mathcal{C}$ be the subset of Z which consists of all Cauchy sequences. Note that the above sequence $a : \mathbb{N} \rightarrow \mathbb{Q}$ defined by $a_0 = 0$ and $a_n = 1$ for all $n > 0$ is Cauchy but it still has no inverse.

Problem 10. (1) Let $a, b \in \mathcal{C}$, show that $a + b, a \cdot b \in \mathcal{C}$.

(2) Let $R = \{(a, b) \in \mathcal{C} \times \mathcal{C} \mid \forall \epsilon > 0, \epsilon \in \mathbb{Q}, \exists N \in \mathbb{N} \text{ s.t. } |a_n - b_n| \leq \epsilon, \forall n \geq N\}$. Show that R is an equivalence relation on $\mathcal{C}$.

(3) Consider $\mathcal{C}/R$, the set of equivalence classes. Show that the above addition and multiplication on Z (and hence on $\mathcal{C}$) can be defined on $\mathcal{C}/R$. In other words, show that the following maps

$$\mathcal{C}/R \times \mathcal{C}/R \rightarrow \mathcal{C}/R$$

$$\bar{a} + \bar{b} := \overline{a + b}$$

and

$$\bar{a} \cdot \bar{b} := \overline{a \cdot b}$$

are both welldefined.

(4) Show that $(\mathcal{C}/R, +, \cdot)$ is a field.

(5) Consider the sequence $a : \mathbb{N} \rightarrow \mathbb{Q}$ defined by $a_0 = 0$ and $a_n = 1$ for all $n > 0$. Show that $\bar{a}$ is invertible in $\mathcal{C}/R$.

The field $\mathcal{C}/R$ is the real number field $\mathbb{R}$. Note there is a natural map $\mathbb{Q} \rightarrow \mathcal{C}/R$ defined by $r \mapsto \bar{\mathbf{r}}$, where $\mathbf{r} : \mathbb{N} \rightarrow \mathbb{Q}$ is the sequence $\mathbf{r}_n = r$ for each $n \in \mathbb{N}$.

Problem 11. (1) Show that the map $\mathbb{Q} \rightarrow \mathcal{C}/R$ defined by $r \mapsto \bar{\mathbf{r}}$ is injective.

(2) Consider the sequence $a : \mathbb{N} \rightarrow \mathbb{Q}$ defined by $a_n = 1 - 10^{-n}$. Show that $a \in \mathcal{C}$ and $\bar{a} = \bar{\mathbf{1}}$. This explains $0.9999 \dots = 1$ in $\mathcal{C}/R$.

One can also define an order on $\mathcal{C}/R$ as follows. A sequence $a \in \mathcal{C}$ is called positive if for every $\epsilon > 0, \epsilon \in \mathbb{Q}$, there exists an N such that $a_n > \epsilon$ (note $a_n, \epsilon \in \mathbb{Q}$ and the sign $>$ means the order in $\mathbb{Q}$.) One checks that if $a \sim b$ and $a > 0$, then $b > 0$. Thus for $\bar{a} \in \mathcal{C}/R$, we can define $\bar{a} > 0$ if $a > 0$. This is independent on the choice of representatives. Then one defines $a > b$ by $a - b > 0$. One then shows that this order is compatible with addition and multiplication in the following sense: (1) if $a > b$, then $a + c > b + c$; (2) If $a > b$ and $c > 0$, then $ac > bc$ for all $a, b, c \in \mathcal{C}/R$. Moreover, $\mathcal{C}/R$ satisfies the least upper bound property: for a non-empty subset $A \subset \mathcal{C}/R$, if A has an upper bound, then it has a least upper bound.